K. VIGNESH, (Ph.D)

AI/ML Researcher & Developer | Generative AI & Computer Vision Specialist | Academic-Industry Bridge

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PROFESSIONAL SUMMARY

AI/ML professional with 16 years of experience in academic institutions, spanning two distinct phases: 7 years as an undergraduate engineering faculty member (2010–2017) teaching core Computer Science subjects, followed by 9+ years (2017–Present) as an AI/ML researcher and developer at the National Institute of Technology, Trichy, building applied solutions in Computer Vision, Natural Language Processing, Deep Learning, Generative AI, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG). PhD in AI/ML with peer-reviewed publications on lecture-video topic segmentation. Hands-on experience designing and shipping end-to-end GenAI applications, chatbots, and multimodal systems using Python, LangChain, LangGraph, Hugging Face, and Google Gemini — combining strong theoretical grounding with practical development skills.

RESEARCH & DEVELOPMENT EXPERIENCE

Altera

May 2025 – May 2026

AI/ML Engineer — PhD Intern

Project: NextGen Triage Debug Tools

- Built an AI-powered failure triage platform that recommends related past HSD records for new failures using E5 transformer embeddings and semantic similarity.
- Designed a Bronze–Silver–Gold data architecture to organize and process failure data across the AI workflow.
- Developed Python backend services integrated with PostgreSQL to serve recommendations to debugging teams.
- **Technologies Used:** E5 Transformer Embeddings, Semantic Similarity, Python, PostgreSQL

National Institute of Technology, Trichy

Aug 2017 – Present

Doctoral Researcher & AI/ML Developer — Designation: Junior Research Fellow (Jul 2020 – Jan 2022)

Transitioned from an instructional role into full-time applied AI/ML research and development, designing and building end-to-end AI systems across NLP, Computer Vision, and Generative AI. Key projects:

Extractive Text Summarization using BERT and Linear Regression

- Designed automated lecture transcript summarization system using BERT and machine learning techniques.
- Implemented extractive summarization pipeline with TextTiling++ for topic-aware content segmentation.
- Improved summary quality and content coverage through feature engineering and regression-based sentence ranking.
- **Technologies Used:** BERT, TextTiling++, Scikit-Learn, Pandas, NumPy, Python

Text-to-Voice Generator using Google Gemini and Hugging Face

- Developed AI-powered text-to-speech platform for generating natural and expressive speech from text.
- Integrated Google Gemini and Hugging Face models for intelligent text processing and voice generation.
- Enhanced speech quality and scalability through transformer-based language processing techniques.
- **Technologies Used:** Google Gemini, Hugging Face, LangChain, Transformers, Python

Handwritten Character Recognition using CNN

- Developed deep learning-based handwritten character recognition system using Convolutional Neural Networks (CNN).
- Implemented image preprocessing and feature extraction techniques for improved character classification accuracy.
- Optimized model performance through data augmentation and hyperparameter tuning strategies.
- **Technologies Used:** CNN, TensorFlow, Keras, OpenCV, Python

Content Generation using Google Generative AI

- Developed AI-driven content generation platform for automated text creation and intelligent content assistance.
- Integrated Google Generative AI and Gemini APIs to generate context-aware and high-quality content.
- Improved content relevance and generation efficiency through prompt engineering and model optimization.
- **Technologies Used:** Google Generative AI, Gemini API, Python

Image-to-Text Description using Google Generative AI

- Designed multimodal image-to-text solution for automatic product description generation.
- Implemented visual understanding pipeline using Gemini Pro Vision and LangChain frameworks.
- Enhanced description accuracy and contextual relevance through advanced vision-language modeling.

- **Technologies Used:** LangChain, Gemini Pro Vision, Google Generative AI, Python

Question Answering from PDF Documents using LangChain

- Designed enterprise document question-answering solution using LangChain, FAISS, Google Embeddings and Retrieval-Augmented Generation (RAG).
- Implemented semantic retrieval pipeline for large PDF repositories.
- Improved answer relevance and reduced hallucinations through optimized retrieval strategies.

Sentiment Analysis, QA and Image Classification

- Developed multi-task AI platform integrating sentiment analysis, question answering, and image classification capabilities.
- Implemented transformer-based NLP models and Vision Transformers (ViT) for accurate text and image understanding.
- Enhanced prediction accuracy and inference efficiency through deep learning model optimization and fine-tuning.
- **Technologies Used:** Transformers, BERT, ViT, PyTorch, Python

Conversational Chatbot using LangGraph and Gemini

- Designed stateful conversational AI assistant using LangGraph, LangChain, and Google Gemini.
- Implemented workflow orchestration, memory management, and context-aware dialogue generation for intelligent interactions.
- Improved user engagement and response quality through advanced prompt engineering and conversation flow optimization.
- **Technologies Used:** LangGraph, LangChain, Gemini, Python

TEACHING EXPERIENCE

Christhu Raj Institute of Computer Applications *June 2010 – July 2014*

Dhanalakshmi Srinivasan Institute of Technology *Aug 2014 – June 2016*

Additional responsibility: ERP Admin, Website Developer

MAM College of Engineering *July 2016 – July 2017*

Delivered undergraduate courses across three engineering colleges, covering: C Programming, C++ and Java Programming, Computer Networks, Operating Systems, and Computer Architecture.

EDUCATION

- Ph.D. — National Institute of Technology, Trichy (Thesis Completed)
- MCA — Jayaram College of Engineering and Technology, Trichy
- B.Sc. Computer Science — Bishop Heber College, Trichy

PUBLICATIONS & RESEARCH

International Journals

- Vignesh, K., and S. R. Balasundaram. "Hierarchical clustering with SBERT and LSH to improve topic boundary segmentation accuracy in lecture videos." International Journal of Information Technology (2025): 1–14. https://doi.org/10.1007/s41870-025-02778-5
- Vignesh, K., and S. R. Balasundaram. "Contrastive Learning-based Multi-scale Neural Processing for Thematic Segmentation of Lecture Video Transcripts." Submitted to Multimedia Tools and Applications (Springer), 2026. (Accepted)

International Conferences

- Vignesh, K., and S. R. Balasundaram. "Thematic Segmentation of Lecture Video Transcripts Using Text Tiling++." 2nd International Conference on Algorithms and Computational Theory for Engineering Applications (ICACTEA 2025).
- Vignesh, K., and S. R. Balasundaram. "Hierarchical Contextual Modeling Based Thematic Segmentation for Handling Discourse Variability in Lecture Video Transcripts." 3rd International Conference on Computing and Machine Learning (CML 2026).
- Vignesh, K., and S. R. Balasundaram. "IRRIGMAN – Irrigation Management Knowledge Sharing and Question Answering Tool Using SBERT and RAG Model." International Conference on Intelligent Computational and Communication Technologies for Sustainable Development (ICCT-SD 2026) — Best Paper Award.

LANGUAGES KNOWN

English, Tamil

REFERENCES

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